SYSTEM ARCHITECTURE, DATABASE PLAN & AI STUDIO PROMPT ENGINEERING FORM

AGRIMATCH-NV: Master Development Blueprint

Adaptive Geospatial and Resilient Intelligence for Multi-Agent Tracking in Crop Hubs Across Nueva Vizcaya

Part 1: Full-Stack System Plan (UI, Hybrid Firebase + SQL, & Multi-Agent Architecture)

This blueprint translates the AGRIMATCH-NV Conceptual Framework and Green Glassmorphism visual references into a production-ready technical specification for Google AI Studio. To satisfy both real-time responsiveness (live counters, GPS pins, instant chat order cards) and structured relational reporting (NVAT market statistics, supply-demand matching, and verifiable completed transactions), the system utilizes a dual-mode hybrid database architecture pairing Firebase with a relational SQL database.

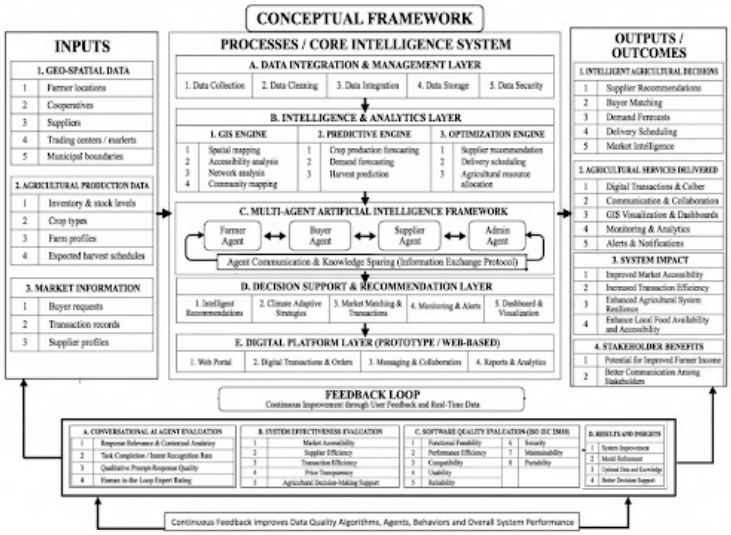


Figure 1: AGRIMATCH-NV Core Intelligence System — Inputs, 5-Agent Framework, Decision Support Layer, and ISO/IEC 25010 Feedback Loop.

1.1 Visual Design System: Green Screen Glassmorphism

Derived directly from the uploaded UI style references and the "Green Screen" color specification, the interface pairs a crisp white splash environment with frosted translucent surfaces, glowing green borders, and deep forest typography.

Token Name	Hex Code	UI & Glassmorphism Application in AGRIMATCH-NV
Mint Whisper	`#F4FFFC`	Splash loading background, primary light surface base, high-contrast text on dark green cards, and clean background behind the animated AGRIMATCH-NV logo.
Frosted Jade	`#91EAAF`	Glassmorphic card borders (`rgba(145, 234, 175, 0.45)`), active GIS radius rings, live pulse indicators, and verified crop badges.
Sprout Lime	`#C3E956`	Primary interactive call-to-action buttons ("Finalize Order & Buy", "Verify Crop"), AGRIBEE companion highlights, and positive price trend bars.
Canopy Olive	`#4D7111`	Secondary navigation pills, active filter states, weather gauge gradients, and role badges (Farmer, Supplier, Buyer, Cooperative, Government).
Deep Forest	`#1F4B2C`	Primary dashboard shell, dark frosted glass panels (`rgba(31, 75, 44, 0.78)` with `backdrop-filter: blur(16px)`), headers, and bottom floating dock.

1.2 Hybrid Database Plan: Firebase + Relational SQL

To ensure zero-latency UI updates on the landing screen alongside rigorous analytical integrity for the Decision Support Layer, data responsibilities are partitioned between Firebase and PostgreSQL/MySQL (via Firebase Data Connect or Supabase):

Subsystem Layer	Technology Engine	Core Tables / Collections & Real-Time Responsibilities
Authentication & Identity Gate	Firebase Authentication	Handles Google OAuth 2.0 Sign-In, Phone Number SMS OTP Verification, Role Claims (`farmer` , `supplier` , `buyer` , `cooperative` , `government`), and the hardcoded Private Admin Credential gate for `russelthegreat1209@gmail.com` .
Live Presence, Chat & Counters	Firebase Cloud Firestore & Realtime DB	Collections: `system_metrics/live_stats` (tracks total site views, currently logged-in sessions, stakeholder role counts, and completed transactions), `private_chats/{threadId}/messages` (stores direct messaging and embedded interactive Order Inquiry Cards), and `community_feed` .
Media & Crop Verification	Firebase Cloud Storage + Gemini Vision	Stores uploaded commodity images in `crop_uploads/{userId}/{postId}` , holding posts in a `PENDING_VERIFICATION` state until verified by the Admin Agent and approved by the Admin account.

Subsystem Layer	Technology Engine	Core Tables / Collections & Real-Time Responsibilities
Geospatial, Inventory & Ledger	Relational SQL (PostgreSQL + PostGIS)	Tables: `users`, `commodities_inventory` (with `GEOGRAPHY(Point, 4326)` for exact radius queries at 10m, 10km, 50km, 100km), `orders_ledger` (tracks dual-confirmation booleans `supplier_sent` and `buyer_received`), and `nvat_price_analytics` (stores AI-parsed NVAT market tables and historical price series).

1.3 Multi-Agent AI & Decision Support Mapping

Every AI requirement in your Conceptual Framework is mapped to a dedicated system instruction and structured JSON schema powered by the Gemini API inside Google AI Studio:

- **AGRIBEE (General Platform Companion):** Guides first-time visitors through the two-step interactive onboarding demo, explains why AGRIMATCH-NV is vital for Nueva Vizcaya farmers, walks users through account creation, and answers general navigation queries across all accounts.
- **a) Farmer Agent (Embedded Exclusively in Farmer + Supplier Accounts):** Manages farm profiles, municipal coordinates, crop types, production schedules, and harvest predictions; organizes crop listings for prospective buyers.
- **b) Buyer Agent (Embedded in Buyer Accounts):** Processes purchase requirements, target kilogram volumes, and expected price thresholds to identify matching crop sources.
- **c) Supplier Agent (Embedded in Farmer/Supplier & Cooperative Accounts):** Tracks real-time commodity stock levels, inventory depletion, and incoming buyer order inquiries.
- **d) Market Agent (Global Intelligence & Weather/NVAT Engine):** Synthesizes PAGASA weather feeds with a 30-day predictive agricultural weather outlook and coordinates regional trading opportunities.
- **e) Admin Agent (Exclusive to Admin Account):** Performs automated authenticity screening on uploaded crop images and executes the **NVAT Text-to-Statistics Pipeline**—converting raw paragraphs or price bulletins from the Nueva Vizcaya Agricultural Terminal into structured statistical tables, commodity variance metrics, and live price charts.

Part 2: Comprehensive, Long-Form Master Prompt for Google AI Studio (Copy & Paste Entire Prompt)

Copy all blocks in Part 2 from start to finish and paste them directly into **Google AI Studio** (alongside your logo image file). Every functional requirement, visual token, database rule, AI agent behavior, and user workflow is explicitly written out below so the AI builder generates the complete system without omitting any detail.

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================================================================================
MASTER SYSTEM PROMPT FOR GOOGLE AI STUDIO: AGRIMATCH-NV FULL-STACK WEB PLATFORM
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ROLE & MISSION: You are an Elite Principal Full-Stack Web Architect, Geospatial Information Systems (GIS) Engineer, Multi-Agent AI Specialist, and Senior UI/UX Product Designer. Your task is to build a complete, functional, production-grade, mobile-and-desktop responsive web application titled "AGRIMATCH-NV" (Adaptive Geospatial and Resilient Intelligence for Multi-Agent Tracking in Crop Hubs Across Nueva Vizcaya). Every screen, button, database listener, map pin, chat workflow, and AI agent described below must be fully interactive, state-driven, and visually polished. ---------------------------------------------------------------------------
-------- MODULE 1: VISUAL DESIGN SYSTEM — "GREEN SCREEN" GLASSMORPHISM & UI REFERENCES ---------------------------------------------------------------------------
--------- Design the entire platform with a high-craft Agricultural SaaS "Green Glassmorphism" aesthetic inspired by modern agri-tech mobile and web dashboards: 1. MANDATORY COLOR PALETTE (Strict Hex Adherence): - Primary Light Surface / Splash Base: #F4FFFC (Mint Whisper) and #FFFFFF (Pure White for the logo stage) - Frosted Glass Accent & Luminous Borders: #91EAAF (Frosted Jade) - High-Contrast Action CTA & Active Highlights: #C3E956 (Sprout Lime) - Secondary Botanical Accent & Role Badges: #4D7111 (Canopy Olive) - Primary Deep Forest Shell, Headers & Dark Glass Cards: #1F4B2C (Deep Forest) 2. GLASSMORPHISM SURFACE SPECIFICATIONS: - Light Frosted Glass Cards: `background: rgba(244, 255, 252, 0.76); backdrop-filter: blur(18px); -webkit-backdrop-filter: blur(18px); border: 1px solid rgba(145, 234, 175, 0.55); border-radius: 22px;` - Dark Forest Frosted Glass Cards: `background: rgba(31, 75, 44, 0.84); backdrop-filter: blur(20px); -webkit-backdrop-filter: blur(20px); border: 1px solid rgba(195, 233, 86, 0.35); border-radius: 22px; color: #F4FFFC;` - Interactive Pill Buttons: Primary buttons use `background: linear-gradient(135deg, #C3E956 0%, #91EAAF 100%); color: #1F4B2C; font-weight: 700; border-radius: 9999px;`. Secondary buttons use `#4D7111` with `#F4FFFC` text. 3. DASHBOARD WIDGET STYLING (Derived from Style References): - Weather Telemetry Widget: Translucent green card displaying current municipal temperature in large typography (e.g., `28°C`), condition pill (`Cloudy` / `Sunny` / `Light Rain`), a horizontal thermal gradient bar spanning `25°C` to `31°C` across hourly markers (`09 AM`, `10 AM`, `12 PM`, `01 PM`, `02 PM`) with a vertical white indicator needle, paired with dual mini-cards for `UV Index` (with an orange/green area sparkline) and `Humidity %` (with a cyan/green area sparkline). - Crop & Market Health Cards: Clean white-and-mint glass cards with a `Live` red pulsing dot badge, percentage KPI readouts (e.g., `94% Crop Quality / Authenticity Score`, `75% Supply Fulfillment`), progress bars (`Days to Harvest`), and vertical rounded capsule bar charts (`Sat`, `Sun`, `Mon`, `Tue`, `Wed`, `Thu`, `Fri`) with `#1F4B2C` and `#4D7111` fills. - Executive KPI Summary Tiles: 2x2 metric grid displaying `Total Shipments / Volume (kg)`, `Total Orders`, `Market Revenue / Value (PHP)`, and `Completed Deliveries`, each with circular pastel icon badges and green/red percentage trend pills (`+1.3% vs last month`), accompanied by a striped green area trend chart and a floating bottom pill navigation dock (`Dashboard`, `GIS Map`, `Marketplace`, `AI Agents`, `Messages`, `Profile`).
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MODULE 2: DUAL-MODE ARCHITECTURE (25-ACCOUNT DEMO MODE + LIVE REAL-TIME MODE) ----
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Implement a persistent, top-bar Environment Mode Toggle that switches seamlessly
between two operational modes: 1. "DEMO WEBSITE MODE" (Pre-Seeded 25 Stakeholder
Accounts + Full AI/GIS Dataset): - Designed for immediate live demonstration,
panel defense, and feature walkthroughs. - Pre-load 25 complete stakeholder
accounts across Nueva Vizcaya (5 accounts per stakeholder role) with realistic
crop inventories, GPS coordinates, verified crop photos, active marketplace posts,
private chat histories, and NVAT price bulletins: * 5 FARMER ACCOUNTS: 1) Mang
Ramon Dela Cruz (Kayapa — Highland Cabbage & Broccoli, `16.3025° N, 120.8614° E`)
2) Aling Teresita Pascual (Bambang — Red Creole Onions & Garlic, `16.3861° N,
121.1065° E`) 3) Danilo Tomas (Kasibu — Satsuma Mandarin Citrus & Pomelo,
`16.3197° N, 121.2878° E`) 4) Elena Magno (Dupax del Sur — Native Ginger & Tomato,
`16.2833° N, 121.1000° E`) 5) Roberto Valera (Solano — Hybrid Yellow Corn & Palay,
`16.5186° N, 121.1819° E`) * 5 SUPPLIER / TRADER ACCOUNTS: 1) NV Highland Agro-
Supply Hub (Bambang NVAT Compound, `16.3912° N, 121.1120° E`) 2) Vizcaya Fresh
Cold-Chain Consolidators (Bayombong, `16.4815° N, 121.1492° E`) 3) Cagayan Valley
Vegetable Trading Post (Aritao, `16.2969° N, 121.0336° E`) 4) Bagabag Agri-
Logistics & Wholesale (Bagabag, `16.6061° N, 121.2519° E`) 5) Santa Fe Mountain
Gate Produce Supply (Santa Fe, `16.1583° N, 120.9389° E`) * 5 BUYER ACCOUNTS: 1)
Metro Balintawak Bulk Buyers Corp. (Institutional Buyer — Seeking 3,000 kg Cabbage
& Tomatoes) 2) Isabela-Vizcaya Supermarket Chain (Retail Buyer — Seeking 1,500 kg
Citrus & Onions) 3) Solano Farm-to-Table Restaurant Group (Hospitality Buyer —
Seeking 400 kg Assorted Highland Greens) 4) Nueva Vizcaya Provincial Hospital
Nutrition Service (Institutional Buyer — Seeking 800 kg Root Crops) 5) Divisoria
Direct Agro-Distributors (Wholesale Buyer — Seeking 5,000 kg Squash, Ginger &
Corn) * 5 COOPERATIVE ACCOUNTS: 1) Kayapa Highlanders Multi-Purpose Cooperative
(KHMPC) 2) Bambang Onion & Vegetable Growers Cooperative (BOVGC) 3) Kasibu Citrus
Producers Cooperative (KCPC) 4) Dupax Agricultural Marketing Cooperative (DAMCO)
5) Nueva Vizcaya Federation of Agro-Cooperatives (NVFAC) * 5 GOVERNMENT / ADMIN
ACCOUNTS: 1) Provincial Agriculture Office — Nueva Vizcaya (PAO-NV Monitor) 2)
NVAT Market Intelligence & Price Monitoring Officer 3) DA Regional Field Office 02
— Agribusiness Coordinator 4) Municipal Agriculture Office — Bayombong/Bambang
Desk 5) MASTER ADMIN ACCOUNT: `russelthegreat1209@gmail.com` (Full System Owner &
Admin Agent Controller) - Include a floating "Demo Quick-Login Switcher" drawer
allowing one-click switching between any of these 25 accounts to test buyer-
supplier chat, dual-button order completion, GIS routing, and Admin verification
in real time. 2. "REAL-TIME PRODUCTION MODE" (Firebase + Relational SQL
Integration): - Connects directly to Firebase Authentication, Cloud Firestore
real-time listeners (`onSnapshot`), Firebase Cloud Storage, and a Relational SQL
backend (PostgreSQL + PostGIS via Firebase Data Connect / Supabase). - Every new
account registration, login session, website visit, commodity post, chat message,
and completed transaction updates the live database and broadcasts instant counter
updates to all connected clients.
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MODULE 3: PHASE 1 — ANIMATED SPLASH LOADING SCREEN (UPON TAPPING/ENTERING LINK) --
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When a user first taps or enters the website URL: 1. Render a full-viewport Splash Loading Screen (`100vw x 100vh`) with a clean, luminous white-to-mint background (`#FFFFFF` blending softly into `#F4FFFC`) so the attached AGRIMATCH-NV Logo (which has a white/transparent background) sits seamlessly in the center without harsh box edges. 2. Animate the AGRIMATCH-NV Logo using a multi-layered CSS keyframe sequence: - Smooth scale-in and gentle floating motion (`@keyframes floatLogo`), - Concentric pulsing emerald and lime rings (`#91EAAF` and `#C3E956`) radiating outward behind the white circular logo badge, - An animated glassmorphic progress bar below the logo that fills from `0%` to `100%` over 2.8 seconds while cycling status messages: * `0% - 35%`: "Initializing AGRIMATCH-NV Core Intelligence System..." * `36% - 70%`: "Connecting PAGASA Weather Telemetry & NVAT Market Feeds..." * `71% - 100%`: "Activating Multi-Agent AI Framework & GIS Spatial Engine..." 3. Include a subtle "Skip Intro" button in the bottom corner and automatically transition to the Public Landing & Intro Screen once hits 100%.
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MODULE 4: PHASE 2 — PUBLIC LANDING & INTRO SCREEN (WITH REAL-TIME TELEMETRY &
GUEST LOCK) ------------------------------------------------------------------
---------- Immediately after the loading screen, display the Public Landing &
Intro Screen containing four live informational tabs and a strict Guest Access
Guard: 1. REAL-TIME TELEMETRY & STAKEHOLDER COUNTERS BAR (Live Database Bound):
Display animated real-time counters with pulsing `#91EAAF` live badges showing: -
Total Website Visitors / Page Views (increments automatically in real time on
every visit) - Currently Logged-In Active Users (live session presence counter) -
Total Completed Transactions (increments whenever a Buyer and Supplier both click
their transaction completion buttons) - Total Commodities / Supplies Posted (total
active listings + total kilograms available across Nueva Vizcaya) - Registered
Accounts Breakdown by Individual Stakeholder Type: * Registered Farmers Count *
Registered Suppliers Count * Registered Buyers Count * Registered Cooperatives
Count * Registered Government / Admin Accounts Count 2. TAB A — REAL-TIME WEATHER
& 1-MONTH AI PREDICTIVE WEATHER OUTLOOK: - Real-Time Weather Panel (Powered by
PAGASA Telemetry + Open-Meteo/Weather API for Nueva Vizcaya): * Municipality
selector covering all 15 municipalities of Nueva Vizcaya (defaulting to Bambang /
Bayombong). * Displays current temperature (`°C`), feels-like, humidity (`%`),
wind speed (`km/h`), UV index, rainfall (`mm`), and the signature glassmorphic
hourly temperature bar (`09 AM` to `02 PM`). - 1-Month (30-Day / 4-Week) AI
Predictive Weather Engine: * Powered by an embedded AI Weather & Climate-Adaptive
Agent that synthesizes PAGASA seasonal projections with local highland/lowland
microclimates. * Displays a 4-Week Predictive Timeline Card (Week 1 to Week 4)
forecasting expected rainfall probability, temperature range, typhoon/monsoon risk
level (`Low`, `Moderate`, `High`), and actionable Crop Advisory Recommendations
(e.g., "Week 3 High Rainfall Alert in Kayapa & Kasibu: Schedule early harvest for
mature highland cabbage and clear drainage canals"). 3. TAB B — OFFICIAL ADMIN
UPDATES, NEWS & NVAT MARKET PRICE STATISTICS: - Displays real-time official
announcements, agricultural bulletins, and subsidies posted by the Admin account.
- Displays the live NVAT (Nueva Vizcaya Agricultural Terminal) Commodity Price
Statistical Tables and Interactive Price Trend Graphs generated by the Admin AI
Agent. 4. TAB C — PUBLIC COMMODITY & SUPPLY OVERVIEW PREVIEW: - Displays a summary
ticker of commodities currently posted in the platform (e.g., Highland Cabbage,
Red Onion, Satsuki Tomato, Ginger, Mandarin Citrus) with aggregate supply volume.
5. STRICT GUEST ACCESS LOCK RULE: - While on the Landing Screen, unauthenticated
Guest users can view the live counters, inspect the Weather Forecast, read Admin
News/NVAT Statistics, and launch the AGRIBEE Interactive Demo. - CRITICAL SECURITY
RULE: If a Guest taps any internal platform navigation tab (such as `Marketplace`,
`GIS Map`, `Private Chat`, `Post Supply`, `Order Commodity`, or taps on a specific
Supplier/Farmer card to enter the website), block navigation and trigger a frosted
glass modal: * Title: "Stakeholder Account Required" * Message: "You are currently
browsing as a Guest. To access the live Marketplace, interactive GIS Map, Private
Chat, and Multi-Agent Decision Support, please sign in or create an AGRIMATCH-NV
account with AGRIBEE." * Buttons: `Start AGRIBEE Onboarding Demo` and `Sign In /
Create Account`.
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MODULE 5: PHASE 3 — AGRIBEE INTERACTIVE ONBOARDING DEMO (STEP 1 & STEP 2) &
AUTHENTICATION ---------------------------------------------------------------
------------- Provide an interactive, guided onboarding experience led by
**AGRIBEE** (the platform's friendly animated green-and-gold AI Bee Companion): 1.
STEP 1 — INTRODUCTION BY AGRIBEE: - Render an interactive glassmorphic walkthrough
stage featuring the animated AGRIBEE avatar with voice-bubble typography: *
Greeting: "Hi! I'm AGRIBEE, your AI companion for AGRIMATCH-NV!" - AGRIBEE
presents 4 interactive feature showcase slides explaining how the platform works
and why it is vital for Nueva Vizcaya farmers and stakeholders: * Slide 1 (The
Problem We Solve): How farmers in Nueva Vizcaya struggle with price opacity,
oversupply bottlenecks at NVAT, unpredictable highland weather, and middleman
markups. * Slide 2 (Direct Buyer-Supplier Matching & GIS Mapping): How GPS radius
search (`10m` to `100km`) and route visualization connect farmers directly with
buyers and cooperatives. * Slide 3 (5-Agent AI Intelligence & NVAT Live
Analytics): How the Farmer, Buyer, Supplier, Market, and Admin AI Agents
coordinate supply, demand, and real-time NVAT price statistics. * Slide 4
(Verified Crops & Protected Transactions): How AI crop image verification and
dual-confirmation chat orders protect every stakeholder. 2. STEP 2 — HOW TO CREATE
AN ACCOUNT (GUIDED BY AGRIBEE): - Clicking "Next: Create Your Account with
AGRIBEE" transitions seamlessly into the guided Registration & Authentication
Wizard where AGRIBEE highlights each step with helper tooltips: * Authentication
Method 1 — Google Authentication: Functional "Sign in with Google" button (OAuth
2.0 popup/redirect flow). * Authentication Method 2 — Phone Number Authentication:
Input Philippine mobile number (`+63 9XX XXX XXXX`), click "Send OTP Code", and
enter the 6-digit SMS verification code. * Authentication Method 3 — Standard
Email & Password Registration/Login. * Stakeholder Role Selection (Mandatory on
Sign-Up): User selects their stakeholder classification card with custom icons: 1)
`Farmer / Supplier` (Produces or supplies agricultural commodities) 2) `Buyer`
(Wholesale trader, retailer, restaurant, institutional buyer, or consumer) 3)
`Cooperative` (Registered agricultural cooperative managing member farms &
logistics) 4) `Government` (DA, PAO, MAO, or NVAT monitoring personnel) * Profile
Completion Fields: Full Name / Organization Name, Municipality in Nueva Vizcaya,
Barangay, Contact Number, and one-tap "Pin My Farm/Business GPS Location". 3.
EXCLUSIVE PRIVATE MASTER ADMIN CREDENTIAL GATE: - There is NO public option to
register an Admin account. - In the Login module, implement an exact credential
check for the Master Admin: * Admin Email: `russelthegreat1209@gmail.com` * Admin
Password: `RusselTheGreat12092009@SagittariusAGRIMATCHNVADMIN` - When—and ONLY
when—a user authenticates with this exact email and password combination, grant
the `MASTER_ADMIN` role token, unlock the **Master Admin Command Center**, and
activate the **Admin AI Agent**.
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MODULE 6: PHASE 4 — MASTER ADMIN COMMAND CENTER & NVAT TEXT-TO-STATISTICS AI AGENT
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When logged in as `russelthegreat1209@gmail.com` (Password: `RusselTheGreat12092009@SagittariusAGRIMATCHNVADMIN`), provide a dedicated real-time Admin Command Center with the following live capabilities: 1. REAL-TIME USER & POST MODERATION SUITE: - View every commodity post, supply listing, and user account across the platform in real time. - Delete Post Action: One-click "Delete Post" button on any marketplace listing or community message (removes it immediately from Firestore/SQL and logs the moderation action). - Ban / Suspend Account Action: Toggle "Ban User Account" on any stakeholder profile (immediately revokes their posting/chat privileges and flags their account as `BANNED`). - Official Community & Landing Page Broadcaster: Compose and publish official updates, price advisories, or weather alerts directly into the Community Chat and the Public Landing Page News Tab. 2. CROP IMAGE AUTHENTICITY VERIFICATION QUEUE (AI VISION + ADMIN APPROVAL): - Whenever a Farmer or Supplier uploads a photo of their agricultural commodity, the post enters a `PENDING_VERIFICATION` queue and is simultaneously analyzed by the Admin AI Agent (Gemini Vision inspection checking for real field/crate photography vs. watermarked/internet stock images, crop type consistency, and visible freshness). - In the Admin Dashboard's "Verify Crops Queue", display each uploaded crop photo, the supplier's details, the AI Vision Authenticity Score (e.g., `96% Authentic Farm Capture`), and two action buttons: * `Verify & Approve Crop`: Marks the commodity as `VERIFIED`, attaches a glowing green "Admin Verified Real Crop" badge (`#91EAAF`) to the listing in the Marketplace and GIS Map, and notifies the supplier. * `Reject Unverified Image`: Hides the listing and prompts the supplier to upload an authentic photo of their actual harvest. 3. NVAT RAW PARAGRAPH-TO-STATISTICS & GRAPHS AI AGENT: - Build a dedicated panel inside the Admin Dashboard titled **"NVAT AI Market Intelligence Processor (Admin Agent)"**. - Provide a large rich-text input area where the Admin pastes raw paragraphs, unstructured text bulletins, or daily market notes from the Nueva Vizcaya Agricultural Terminal (NVAT) (for example: *"NVAT Bambang Daily Price Bulletin: Highland cabbage inflows reached 42 metric tons today with prevailing wholesale prices dropping from Php 38/kg to Php 32/kg due to peak harvest in Kayapa. Red onions remained steady at Php 95/kg to Php 110/kg with 18 tons traded, while Satsuki tomatoes surged to Php 55/kg amid strong Metro Manila buyer demand..."*). - When the Admin clicks **"Process NVAT Text with AI Agent & Publish"**, the Admin Agent automatically parses the unstructured paragraph into a structured JSON dataset and immediately generates: * A formatted **Statistical Market Data Table** containing: `Commodity Name`, `Category`, `Minimum Price (PHP/kg)`, `Maximum Price (PHP/kg)`, `Prevailing Price (PHP/kg)`, `Previous Price (PHP/kg)`, `Price Variance (%)`, `Estimated Volume Traded (kg/tons)`, and `Market Trend Status` (`Surging`, `Stable`, `Declining`), * **Interactive Statistical Charts**: (1) Prevailing vs. Previous Price Comparison Bar Graph, (2) Commodity Supply Volume Share Donut/Bar Chart, and (3) 7-Day Historical + Projected Price Trend Line Graph, * **Executive AI Market Summary**: Bullet-point takeaways for farmers and buyers. - Clicking **"Publish to Live Price Updates"** saves the structured record to the database and pushes the statistical tables and graphs live to the Public Landing Page and all Stakeholder Dashboards in real time.
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MODULE 7: PHASE 5 — INTERACTIVE GIS ENGINE, PROXIMITY RADIUS & 2-TAP PIN ROUTING -
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Once a registered user logs into their AGRIMATCH-NV Dashboard, provide a full-
featured **Geographic Information System (GIS) Integrated Map**: 1. SPATIAL MAP
CANVAS & REAL-TIME GPS ACQUISITION: - Center the interactive map on Nueva Vizcaya
(`16.4830° N, 121.1500° E`) with satellite, terrain, and dark/green glassmorphic
street layer toggles. - Automatically detect the logged-in user's live GPS
coordinates (`navigator.geolocation`) or allow them to drag their "My Location"
marker across any municipality in Nueva Vizcaya. 2. PROXIMITY RADIUS PARAMETER
FILTER (`10m`, `10km`, `50km`, `100km`): - Provide quick-select glassmorphic
radius buttons (`10m`, `10km`, `50km`, `100km`, `All NV`) plus a fine-grained
custom radius input. - When a radius is selected: * Draw a translucent `#91EAAF` /
`#C3E956` geodesic circle around the user's GPS pin on the map, * Compute exact
spatial distance using the Haversine formula / PostGIS `ST_DWithin` between the
user and all posted commodities/stakeholders, and * Filter the visible map pins
and sidebar list to display only Farmers, Suppliers, Cooperatives, and Buyers
within that exact radius. 3. 2-TAP INTERACTIVE COMMODITY PIN STATE MACHINE: -
Every user who has posted commodities in the Marketplace appears as a custom
color-coded pin on the GIS Map (with a crop icon and price tag pill). - **FIRST
TAP ON A PIN (Preview State):** * Opens a floating glassmorphic Pin Preview
Popover anchored to the marker showing: Stakeholder Name, Role Badge (`Farmer`,
`Supplier`, `Cooperative`), Verified Crop Badge, exact distance from the user
(e.g., `4.2 km away`), and a summary list of their active commodities with
available kg and PHP/kg prices, plus a prompt: *"Tap pin or card again to open
full profile & storefront"*. - **SECOND TAP ON THE SAME PIN OR CARD (Full Profile
& Storefront State):** * Opens the stakeholder's complete Profile & Storefront
View displaying: Full Bio, Municipality/Barangay, Verified Contact Number,
Farm/Facility Photos, Complete Catalog of Posted Commodities (with "Order / Buy
Now" buttons on each item), Expected Harvest Schedules, and a direct **"Chat
Privately"** button. 4. LIVE ROUTE VISUALIZATION (FROM USER GPS TO SELECTED PIN):
- On any selected map pin or profile, clicking **"Show Route to Supplier/Farmer"**
renders a glowing `#C3E956` / `#1F4B2C` road route polyline connecting the user's
current GPS coordinates to the target pin, complete with a floating route
telemetry banner showing **Total Distance (km)**, **Estimated Driving Time
(mins)**, and **Origin-to-Destination Municipal Path**.
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MODULE 8: PHASE 6 — MARKETPLACE ORDERING, PRIVATE CHAT & DUAL-BUTTON TRANSACTION
COMPLETION ------------------------------------------------------------------------
--------- Build the complete end-to-end agricultural e-commerce and negotiation
pipeline linking the Marketplace, Inquiries, Private Chat, and Live Database
Counters: 1. MARKETPLACE COMMODITY SELECTION & ORDER INQUIRY CREATION: - When a
user selects a commodity or supply listing in the Marketplace (or from a GIS Map
pin profile), open the **"Configure & Finalize Order"** Glassmorphic Modal
displaying: * Commodity Name, Crop Category, Verified Status Badge, Supplier Name,
Location, and Available Stock (in kg). * Interactive Order Inputs: 1) **Quantity
to Buy (in Kilograms / kg):** Numeric input and slider (validated against
available inventory). 2) **Supplier's Listed Price (PHP/kg):** Read-only reference
price. 3) **Your Expected / Proposed Price (PHP/kg):** Editable price field
allowing the buyer to accept the listed price or propose a negotiated per-kilogram
price. 4) **Auto-Calculated Total Order Value (PHP):** `Quantity (kg) × Proposed
Price (PHP/kg)`. - **TAPPING "FINALIZE ORDER & BUY":** * Immediately creates an
active Order Record with status `INQUIRY_SUBMITTED`, * Adds the order to the
user's **"My Inquiries & Orders"** tab, * Dispatches a real-time push/in-app
notification to the Supplier (`"New Order Inquiry from [Buyer Name] for [X] kg of
[Commodity] at PHP [Y]/kg"`), and * Automatically opens a dedicated **Private Chat
Thread** between the Buyer and the Supplier containing a pinned, interactive
**Order & Fulfillment Card**. 2. PRIVATE CHAT NEGOTIATION & LOGISTICS SETTLEMENT:
- Inside the Private Chat between the Buyer and Supplier: * Both users can
exchange real-time text messages, photos, and location pins, and view each other's
verified profile details and contact number. * The embedded interactive Order Card
allows both parties to select and lock in: 1) **Delivery / Fulfillment Mode:**
`Buyer Farm Pickup`, `Supplier Direct Delivery`, `NVAT Bambang Drop-off Bay`, or
`Cooperative Shared Hauling`. 2) **Meeting / Delivery Location:** Specific address
or GPS pin. 3) **Payment Option:** `Cash on Delivery (COD)`, `GCash / Maya Instant
Transfer`, `LandBank / Bank Transfer`, or `Cooperative Escrow Account`. 4) **Final
Agreed Price per kg & Total Amount (PHP).** 3. DUAL-HANDSHAKE TRANSACTION
COMPLETION (`ORDER SENT` + `ORDER RECEIVED`): - At the bottom of the interactive
Order Card inside the Private Chat (and mirrored in the Inquiries tab), render two
synchronized, role-specific confirmation buttons: * **Button 1 (Supplier Side):**
`Mark as Order Sent / Dispatched` - When clicked by the Supplier, sets
`supplier_confirmed_sent = true` with a timestamp, updates the badge to `Order
Dispatched by Supplier`, and sends an instant notification to the Buyer: *"The
supplier has dispatched your order! Please click 'Order Received' once
delivered."* * **Button 2 (Buyer Side):** `Mark as Order Received` - When clicked
by the Buyer, sets `buyer_confirmed_received = true` with a timestamp, updates the
badge to `Order Received by Buyer`, and sends an instant notification to the
Supplier. - **ATOMIC COMPLETION & REAL-TIME COUNTER UPDATE RULE:** * The moment
**BOTH** buttons have been clicked (`supplier_confirmed_sent === true` AND
`buyer_confirmed_received === true`), the system automatically: 1) Transitions the
transaction status to `COMPLETED`, 2) Writes the permanent transaction record
(Order ID, Buyer ID, Supplier ID, Commodity, Kilograms, Final PHP Amount, Delivery
Mode, Payment Method, Completion Timestamp) into Firebase Firestore AND the SQL
`orders_ledger` table, 3) Deducts the purchased kilograms from the Supplier's
active commodity inventory, and 4) **Immediately increments the "Completed
Transactions" counter by +1 on the Public Landing Screen and Dashboard Analytics
in real time.**
```

```
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MODULE 9: PHASE 7 — MULTI-AGENT AI FRAMEWORK, ROLE-SPECIFIC AGENTS & DECISION
SUPPORT LAYER ---------------------------------------------------------------
------------ Implement the complete **Multi-Agent Artificial Intelligence
Framework** and **Agricultural Supply Chain Decision Support Layer** directly from
the Conceptual Framework: 1. GENERAL AI COMPANION — **AGRIBEE** (Available Across
All Screens): - Accessible via a floating glassmorphic bee launcher in the bottom-
right corner of every page. - Assists with onboarding, platform navigation,
voice/text Q&A in English, Tagalog, or Ilocano, and routes specialized analytical
requests to the 5 domain agents below. 2. THE 5 INTERCONNECTED DOMAIN AGENTS (With
Strict Account Role Binding): - **a) FARMER AGENT (Embedded EXCLUSIVELY in Farmer
+ Supplier Accounts):** * Role Visibility: Render the "Farmer AI Agent Console"
tab ONLY when the logged-in user's role is `Farmer` or `Supplier`. * Core
Functions: Manages farmer profiles, farm municipal/barangay locations, crop
varieties, planting-to-harvest production status, yield estimation, and available
agricultural commodities; automatically formats and organizes farm supply records
to maximize visibility and appeal to matching buyers. - **b) BUYER AGENT (Embedded
in Buyer Accounts):** * Core Functions: Processes buyer profiles, target commodity
specifications, volume requirements (kg), budget ceilings, and purchase requests;
scans active Nueva Vizcaya farm and supplier inventories to identify and rank
optimal sources that match the buyer's exact criteria. - **c) SUPPLIER AGENT
(Embedded in Supplier, Farmer & Cooperative Accounts):** * Core Functions: Manages
supplier profiles, real-time warehouse/farm commodity availability, stock
depletion alerts, restocking schedules, and supply records; matches available
inventory against incoming buyer demand signals. - **d) MARKET AGENT (Global
Market Coordination Engine):** * Core Functions: Coordinates regional agricultural
market information, aggregate supply-demand ratios across Nueva Vizcaya
municipalities, NVAT price trends, and emerging trading opportunities; facilitates
market connections among farmers, suppliers, cooperatives, and buyers. - **e)
ADMIN AGENT (Embedded EXCLUSIVELY in Master Admin Account
`russelthegreat1209@gmail.com`):** * Core Functions: Oversees administrative
governance, system information integrity, user access monitoring, AI Vision crop
photo verification, and the NVAT Raw Text-to-Statistics & Graphs conversion
pipeline. 3. AGENT COMMUNICATION & KNOWLEDGE SHARING PROTOCOL (Information
Exchange Visualizer): - Include an interactive **"Multi-Agent Coordination &
Knowledge Sharing Hub"** view that visualizes live JSON message packets flowing
between the `Farmer Agent ↔ Buyer Agent ↔ Supplier Agent ↔ Market Agent ↔ Admin
Agent`. - Explicitly display a governance notice stating: *"Decision-Support
Governance: The Multi-Agent Framework operates as an intelligent software
coordination mechanism. All recommendations and outputs are generated from
available platform data and system rules and remain subject to stakeholder
validation."* 4. AGRICULTURAL SUPPLY CHAIN DECISION SUPPORT LAYER (5 Actionable
Output Modules): Render 5 dedicated Decision Support sub-tabs powered by the
interconnected agents: - **(a) Intelligent Supplier Recommendations:** Ranks and
recommends top suppliers/farmers for any buyer query using a composite weighted
score of Price Competitiveness (35%), Proximity Distance in km (30%), Verified
Crop Quality (20%), and Inventory Availability (15%). - **(b) Buyer-Supplier
Matching Engine:** Automated two-way matching board comparing open Buyer Requests
(demand) against active Farmer/Supplier Listings (supply) and generating one-click
"Initiate Matched Trade" buttons. - **(c) Agricultural Inventory & Availability
Coordination:** Interactive municipal heatmap and harvest calendar showing current
stock levels (kg), upcoming harvest dates over the next 30–60 days, and regional
surplus/shortage alerts. - **(d) Transaction & Delivery Scheduling Support:**
```

Part 3: Step-by-Step Guide to Building & Deploying the Two Website Modes

Follow this sequential engineering workflow to take your project from Google AI Studio to a fully deployed, publicly accessible web application operating both your **25-Account Demo Website** and your **Real-Time Production Website**.

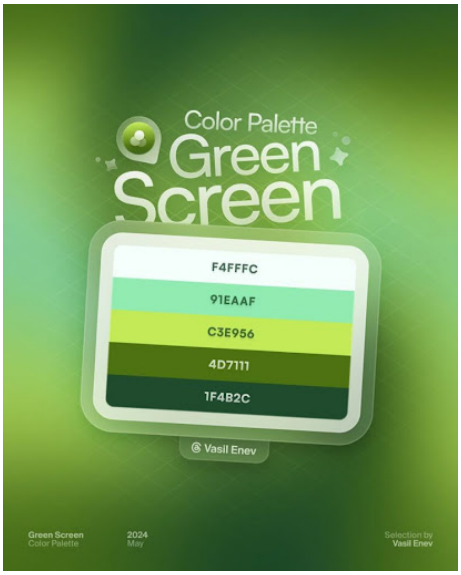


Figure 2: Reference "Green Screen" Color Palette (`#F4FFFC` , `#91EAAF` , `#C3E956` , `#4D7111` , `#1F4B2C`) applied across the glassmorphic interface.

Step 1: Generate the Core Application in Google AI Studio

1. Open **Google AI Studio** (`aistudio.google.com`) and select the **Build / Web App** environment with Gemini 2.5 Pro or Gemini 3 Flash/Pro selected.
2. Upload your transparent **AGRIMATCH-NV Logo**, the **Conceptual Framework** (`https://lh3.googleusercontent.com/gg-dl/AHiVA1miodZcT_zkUVVjmZk762238-Ql_VgnHHoAQuuvbOitofB7E8zqLJGlLk9i8Is6yy3EE4jgfqRV9FkdWIhkAwkGGzHYN80d2Jhvfm6UJEtWLJgTdcBGT6j_2Kkbu6KxmTNfDfy3SyPm573qBNCRTrAZk8VTJF723ILdEMcrBzrC_58n6w`), and the **Green Screen Palette** (`https://lh3.googleusercontent.com/gg-dl/AHiVA1kUnHbkzkHOsbT1UDVT09erKnDJAjM7noI-yGDESSiw6npAJzAjfobOPHEbXAkYEnTkVmS2958Ul7hcQex81tLkpIJUiOldjH-4IV-JjhuuTdWHMLNS87I9KUANvZxutA3q0BLX0n7RSvtmrbUJ1es4WfSq7DzvooeZuHfyej1qBu7kow`) into the prompt context attachments.
3. Paste the **Master Prompt from Part 2** into the prompt box and run generation. Verify that the splash animation, guest lock modal, AGRIBEE onboarding tour, and private admin login (`russelthegreat1209@gmail.com`) render properly in the interactive preview.

Step 2: Configure the 25-Account Demo Environment (For Panel Defense & Testing)

To ensure evaluators and panelists can immediately see the multi-agent intelligence, GIS routing, and NVAT graphs in action without waiting for external users to sign up, seed the Demo Mode state with 5 pre-configured profiles per stakeholder category:

Stakeholder Role	5 Demo Accounts (IDs)	Pre-Loaded Nueva Vizcaya Demo Dataset for AI Agents
1. Farmers (5)	`farmer_01` to `farmer_05`	Located in Kayapa, Bambang, Dupax del Sur, Solano, and Kasibu. Pre-loaded with Highland Cabbage (1,200 kg), Red Onion (850 kg), Satsuki Tomatoes (600 kg), Ginger (500 kg), and Mandarin Citrus (950 kg) with GPS pins and harvest dates.
2. Suppliers (5)	`supplier_01` to `supplier_05`	Located around NVAT Bambang, Bayombong, and Aritao. Pre-loaded with wholesale consolidated vegetable crates, cold-storage inventory schedules, and active delivery trucks.
3. Buyers (5)	`buyer_01` to `buyer_05`	Metro Manila/Isabela wholesale traders, local restaurant chains, and public market vendors with active purchase requests, target PHP/kg budgets, and open chat inquiries.
4. Cooperatives (5)	`coop_01` to `coop_05`	Multi-purpose agricultural cooperatives across Nueva Vizcaya managing aggregated member inventories, shared hauling schedules, and cooperative escrow options.
5. Gov / Admin (5)	`gov_01` to `gov_04` + Master Admin	Provincial/Municipal Agriculture Office monitors, NVAT market price bulletin officers, plus your Master Admin account (`russelthegreat1209@gmail.com`).

Step 3: Connect Firebase & Relational SQL for Real-Time Production Mode

1. **Create a Firebase Project:** Go to `console.firebase.google.com`, create a project named `agrimatch-nv-prod`, and enable **Authentication** (turn on *Google Sign-In*, *Phone SMS Auth*, and *Email/Password*).
2. **Provision Firestore & Cloud Storage:** Create a Firestore database in production mode with real-time snapshot listeners (`onSnapshot`) bound to `system_metrics/live_stats` and `private_chats` so that whenever any user logs in, posts a crop, or clicks both "Order Sent" and "Order Received", every connected browser updates its Landing Screen counters within milliseconds.
3. **Provision Relational SQL (Firebase Data Connect / Supabase PostgreSQL):** Create the relational tables (`users` , `commodities_inventory` , `orders_ledger` , `nvat_price_analytics`) and enable the `postgis` extension to support instant spatial radius queries (`ST_DWithin` for 10m, 10km, 50km, and 100km).
4. **Connect API Keys & Deploy:** Export your web app from Google AI Studio (or deploy directly via **Firebase Hosting / Google Cloud Run**), add your `GEMINI_API_KEY` , `FIREBASE_CONFIG` , and `GOOGLE_MAPS_API_KEY` to your environment variables, and run `firebase deploy` to publish your live link.